Probability & Statistics

$$B(a,b) = \int_0^1 x^{a-1} (1-x)^{b-1} \quad \text{for } a>0,\ b>0$$

$$= \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}$$

<u>Define:</u>

X to be cts r.v. with the pdf

$$f_X(x) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} x^{a-1} (1-x)^{b-1} \quad 0 \le x \le 1$$

$$= 0 \quad \text{o.w.}$$

for $a>0$ and $b>0$.

Then we say X follows Beta density with parameters a & b. $X \sim \text{Beta}(a,b)$

$$E(X) = \frac{a}{a+b}$$

$$var(X) = \frac{ab}{(a+b)^2(a+b+1)} = E(X^2) - [E(X)]^2$$

Support of $Beta(a,b)$ is $(0,1)$.

Weibull distribution.

A random variable X is said to follow Weibull distribution if its pdf is given by

$$f_X(x) = \frac{\beta}{\delta}\left(\frac{x-\gamma}{\delta}\right)^{\beta-1} \exp\left[-\left(\frac{x-\gamma}{\delta}\right)^{\beta}\right], \quad x \geq \gamma$$

$$= 0 \quad \text{o.w.}$$

where $\gamma \in \mathbb{R}$: location, $\delta > 0$ scale parameter

$\beta > 0$ shape parameter.

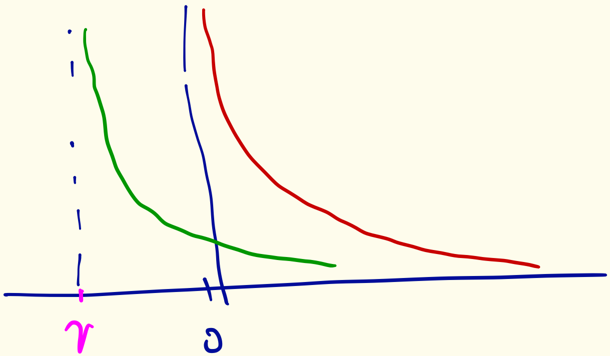
$$E(X) = \gamma + \delta\,\Gamma\left(1+\frac{1}{\beta}\right)$$

$$var(X) = \delta^2\left\{\Gamma\left(1+\frac{2}{\beta}\right) - \left[\Gamma\left(1+\frac{1}{\beta}\right)\right]^2\right\}$$

CDF

$$F_X(x) = 1 - \exp\left[-\left(\frac{x-\gamma}{\delta}\right)^{\beta}\right], \quad x \geq \gamma$$

$$= 0 \quad \text{o.w.}$$



Log normal density:

Consider a random variable X with range $R_X = (0, \infty)$ where $Y = \ln X$ is a normally distributed random variable with mean $E(Y) = \mu_Y$ and variance $var(Y) = \sigma_Y^2$.

then the density of X is given by,

$$f_x(x) = \frac{1}{x\sigma_Y\sqrt{2\pi}} e^{-\frac{1}{2}\left[\frac{(\ln x - \mu_Y)}{\sigma_Y}\right]^2} \qquad x > 0$$

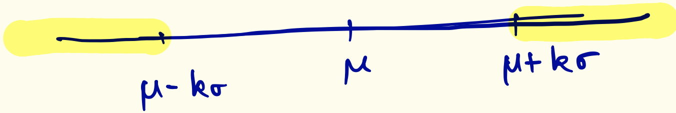
$$= 0 \qquad \text{o.w.}$$

$E(X) = \mu_X = e^{\mu_Y + \frac{1}{2}\sigma_Y^2}$; $var(X) = \mu_X^2 \left(e^{\sigma_Y^2} - 1\right)$

Chebyshev's inequality:

Thm: Let X be a random variable and let k be some positive number.

Then $$P(|X-\mu| \geq k\sigma) \leq \frac{1}{k^2}$$



Pf:

X : random variable

mean of X = $E(X)$

median of X : a pt. $m \in \mathbb{R}$ s.t.

$F_X(m) = 0.5$

quartiles of X : a pt. $q_i \in \mathbb{R}$ s.t.

$F_X(q_i) = \frac{i}{4}$ $\quad i = 1, 2, 3, 4$

percentiles of X : a pt. $p_i \in \mathbb{R}$ s.t.

$F_X(p_i) = \frac{i}{100}$ $\quad i = 1, 2, \ldots, 99$

2^{nd} quartile = 50^{th} percentile = median mode of X $= \underset{x \in R_X}{\arg\max} \; f_X(x)$

MSE - mean square error

$E(X - a)^2$